

Wireless and Mobile Computing

Lecture 1
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Wireless?

- ❑ In communication engineering, “wireless” refers to communication without the use of wires over any distance from a millimeter to vast deep space, beyond our solar system.
 - ❖ Is Wireless a Game changer—Or is It just a Fad?
- ✓ The word “wireless” is often used in the context of a smart phone. However, there is vast area of wireless beyond the smart phone.
- ✓ Even within the smart phone, there are two distinct segments that access the Internet using (a) the cellular network and (b) the WiFi network (which is part of LAN).
- ✓ Use of the word “wireless” to mean both networks and the assumption that these two are similar and interchangeable is quite a common notion.
- ✓ But the cellular network is much larger and independent of the wired LAN network.
- ✓ WiFi is a 100 m “over the air” extension for a wired LAN (IP network). For example, WiFi in a bus or a train can only work if it gets connected to Internet through a cellular or satellite network.
- ✓ Therefore, cellular is the essential wireless network and supports devices inside any building as well as outside such as on the street and beyond (land, water, and air).

- ❑ The concept of cellular started in the late 1940s when covering large geographic areas with cells was conceived.
- ❑ Almost 30 years later, in the 1970s, the first generation or 1G was tried in various parts of the world and national standards emerged slowly in the 1980s.
- ❑ With major improvements in technology, such as chip fabrication and dramatic improvements in reducing power consumption (size of battery), wireless bounced back as a game changer only in the past 20 years.
- ❑ It took center stage of communication networks in the beginning of twenty-first century. Everyone today, affirms the use of cell phone and its impact on society. Figure 1.1 shows a range of cell phones that started in 1973 and now all the way to 5G showing dramatic reduction in size, yet major improvement in performance.
 - ✓ Early versions were bulky, consumed a lot of power. If used in cars, it was even necessary to use car battery and attach the portable unit next to the driver.
 - ✓ Figure 1.2 shows a typical mobile unit during 1G which was so bulky that rested in a compartment within the vehicle. The antenna was long, and it was necessary to mount it on roof of the vehicle, for it to work properly.
 - ✓ The 1G phones typically weighed 5 kg, emitting 5 W but could just make a voice call. It took almost 10 h to charge the battery that supported only 30-min of voice call. In contrast, 4G smart phone today weighs about $\frac{1}{4}$ Kg and emits $\frac{1}{5}$ W while performing innumerable tasks in addition to phone call and Internet access. The change in three decades is quite dramatic.

Fig. 1.1 Evolution of cell phones from 1 to 4G



Fig. 1.2 Portable 1G cell phone used in cars



What Are the Wireless Technologies?

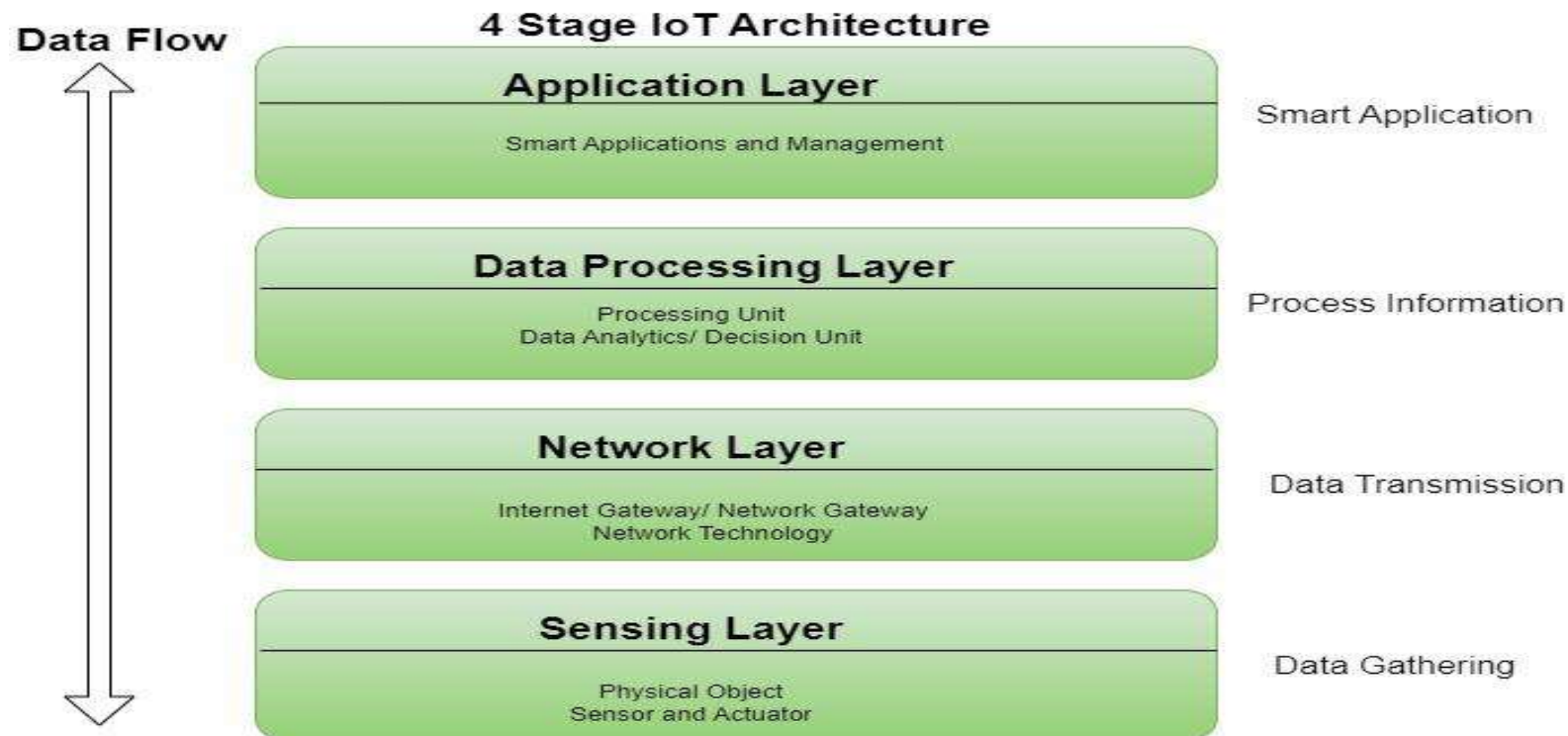
- ❑ Wireless technologies could be classified based on distance they range from the closest (near field) that operates within a few centimeters, to the farthest which is into deep outer space Voyager which is already beyond our solar system continues to send us signals.
- ❑ Wireless communication includes all those operating in air (earth's troposphere), to satellites and beyond that operate mostly in vacuum.
 - ✓ Near field and short distance: (100 m or less): RFID (Radio Frequency Identification), Blue tooth, WiFi, ZigBee, wireless microphones, Beacons/remote starters in cars (fixed and mobile), garage door openers and others.
 - ✓ Medium distance with mobility (between 100 m and 10 km): Broadcast systems (Radio and TV), cellular systems, land mobile systems (public safety), police radar, airport instrument landing systems, port traffic and docking systems, marine radio (boat to boat and beacons), personal beacons in races (running, bicycle, cars, marathons), Amateur Band radio.
 - ✓ Long-distance communication (between 10 and 100 km): Doppler radar, short wave radio (radio broadcast, Amateur Band international and others), marine ship-to-ship HF band (international waters). Long-haul microwave links.

What Are the Wireless Technologies?

- ✓ Very long distances and regional services (between 100 and 1000 km): Satellite services such as satellite digital radio broadcast, satellite communication links (regional spot beam within a region), emergency communication links to support disaster zones, public safety FirstNet, cellular systems within an entire region, ground radar, and navigation support for commercial aircraft.
- ✓ Deep space/international communication (beyond 1000 km): Communication during space missions, Astro Physics and radio telescopes, missions such as Pioneer, Mercury, Voyager, and others. Intercontinental phone/data calls travelling through major satellite earth station links. These include calls to aircraft, ships, and remote regions.

Internet of Things (IoT)

- ❑ There is a whole world of wireless devices that can fill the void of sensor networks, in general referred to as “machine to machine communication”. These are devices that work in locations like at home, outside, in industry and all around provide information based on context, just a few times a day.
- ❑ To standardize and visualize technology based on wireless sensor network and IP network is important to note that data sent by “Things” each time is no more than a few hundred bytes.
- ❑ The architecture of IoT is divided into 4 different layers i.e. Sensing Layer, Network Layer, Data processing Layer, and Application Layer.



- ❑ Sensing Layer: The sensing layer is responsible for collecting data from different sources. This layer includes [sensors](#) and [actuators](#) that are placed in the environment to gather information about temperature, humidity, light, sound, and other physical parameters.
- ❑ Network Layer: The network layer is responsible for providing communication and connectivity between devices in the IoT system. It includes protocols and technologies that enable devices to connect and communicate with each other and with the wider internet. Examples of network technologies that are commonly used in IoT include [WiFi](#), Bluetooth, Zigbee, and cellular networks such as 4G and [5G technology](#). Additionally, the network layer may include [gateways](#) and [routers](#) that act as intermediaries between devices and the wider internet.
- ❑ Data processing Layer: The data processing layer refers to the software and hardware components that are responsible for collecting, analyzing, and interpreting data from IoT devices. This layer is responsible for receiving raw data from the devices, processing it, and making it available for further analysis or action.
- ❑ Application Layer: The application layer interacts directly with the end-user. It is responsible for providing user-friendly interfaces and functionalities that enable users to access and control IoT devices. This layer includes various software and applications such as mobile apps, web portals, and other user interfaces that are designed to interact with the underlying IoT infrastructure.

Background to Wireless

1. Radio Wave Propagation—Basics In practice, the term “radio wave” is applied to those used for wireless communication that quite often limits the range. Therefore, the term does not include the lowest frequencies, nor does it include light or highest frequencies. However, the term “radiation” applies to all of them. Briefly, the accepted range of frequencies for radio wave include the range 300 kHz–300 GHz. Basically, all electromagnetic waves are governed by the fundamental equation: $C = \lambda \cdot f$ where C = Speed of light. At about 3×10^8 m/s, this is a well-known constant described in Physics; this , Equation can be summed up in words as “the product of frequency and wavelength is always constant and is expressed by the speed of light”. This is perhaps the most important fundamental equation that applies to all electro-magnetic radiation including radio waves.

➤ The following are some of the properties of propagation:

❑ **Line-of-sight propagation** is transmission of signals without refraction, diffraction, or scattering between the transmitter and the receiver. However, even in such an ideal scenario, transmission losses do occur. Signal strength in free space decreases as the square of the distance from the transmitter.

Example 1.4

A transmitter sends a signal that has the strength of $9 \mu\text{W}/\text{cm}^2$ at a distance of 500 m. Assuming free-space propagation in line of sight, calculate the signal strength at 1500 m.

Solution:

The strength of the transmitted signal is inversely proportional to the square of the distance from the transmitter.

Therefore, the strength of the signal at 1500 m $= (500/1500)^2 \times 9 \mu\text{W}/\text{cm}^2 = 1 \mu\text{W}/\text{cm}^2$.

- ❑ **Signal strength also decreases** due to attenuation when obstacles in the path of the signal are greater in size than the wavelength of the signal. A few examples of attenuation of signals are as follows: (a) If an FM radio transmitter sends out a 90-MHz (wavelength = 3.3 m) FM band signal, then the signal will be attenuated by objects of size 10 m and above (size much greater than wavelength).
- ❑ **A signal scatters** when it encounters an obstacle of size equal to or less than the wavelength. For example, a GSM signal, about 33 cm in wavelength, is scattered by an object of 30 cm or less. Figure 1.5(a) shows the scattering of a transmitted signal. Only a small part of the scattered signal reaches the receiver.
- ❑ **A signal bends as a result of diffraction** from the edges of an obstacle of size equal to or less than the wavelength. For example, a GSM signal of wavelength 33 cm will diffract from an object of 33 cm or less. Figure 1.5(b) shows a diffracted signal. A diffracted signal may or may not reach its destination; it depends on the geometry of the obstacle and the separation between the object, the receiver, and the transmitter.

❑ **The signal may also be reflected** from the surface of an obstacle, the earth's surface, or a water body of size greater than the wavelength of the signal. For example, if a transmitter sends out a GSM 900 MHz ($\lambda = 33$ cm) signal, then the transmitter signal reflects from an object of size 10 m and above (much greater than λ). The reflected signal suffers a delay in reaching its destination. Figure 1.5(c) shows the reflection and the delay. The delay is more pronounced in case of multi-hop paths. Delayed signals have distorted waveforms and cause misrepresentation of information encoded in the signal. There are digital signal processing (DSP) techniques to eliminate the distortions due to delays from direct and multiple paths so that the original signal can be recovered. The delay in the reflected signal with respect to the original direct signal is given as follows:

$$\textbf{Delay} = (t_{indirect} - t_{direct})$$

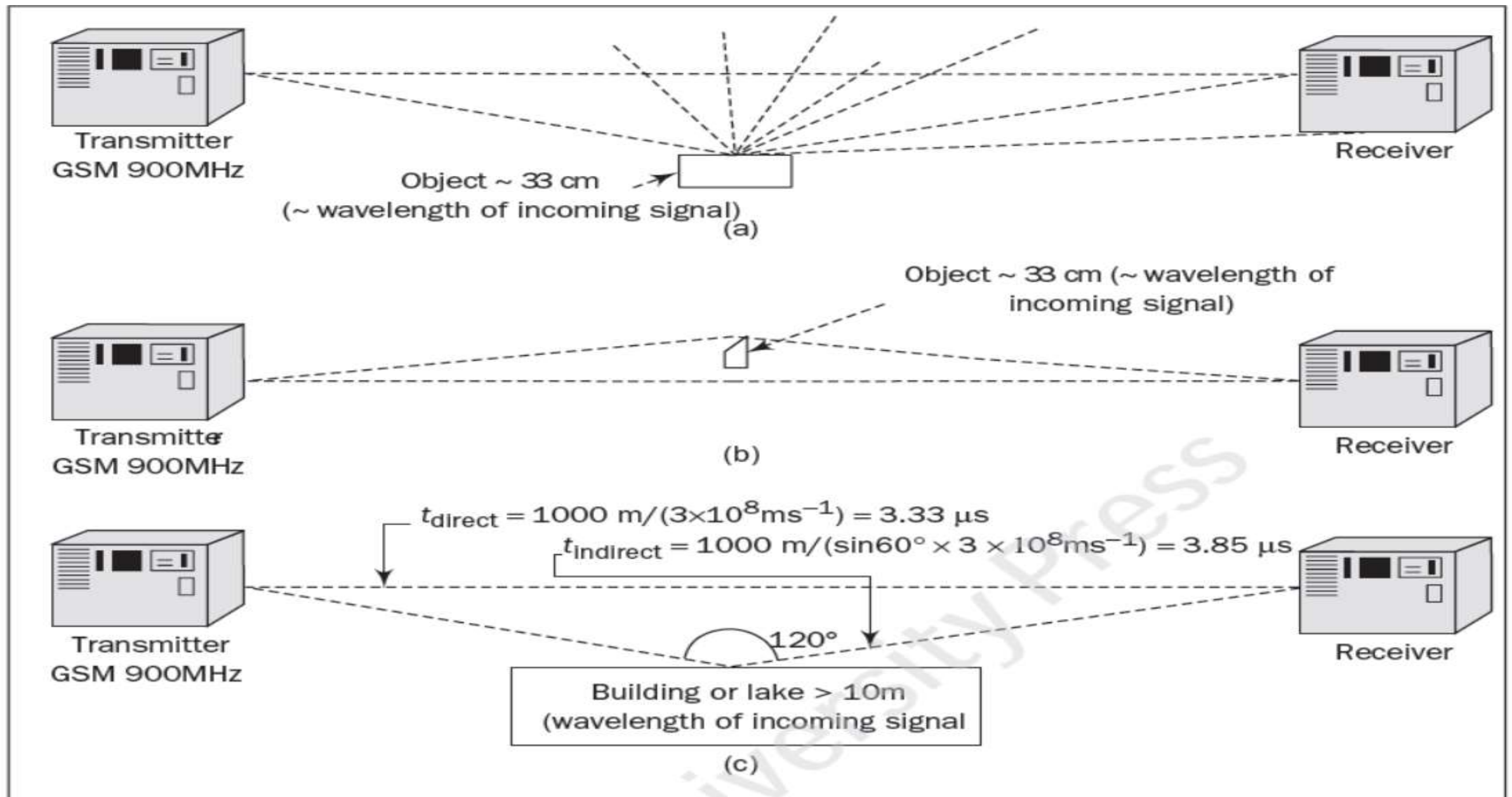


Fig. 1.5 (a) Scattering of signal, (b) Diffraction of signal, and (c) Reflected and direct signals from a 900 MHz transmitter and calculation of delay

Example 1.5

A receiver receives two signals, one directly in line-of-sight and the other after a reflection of 120° from a transmitter at a distance of 1000 m [Fig. 1.5(c)]. Calculate the delay in the reflected signal with respect to the direct signal.

Solution:

$$\text{Direct path time, } t_{\text{direct}} = \frac{1000 \text{ m}}{3 \times 10^8 \text{ ms}^{-1}} = 3.33 \mu\text{s}$$

$$\text{Reflected path time, } t_{\text{indirect}} = \frac{1000 \text{ m}}{\sin(120^\circ/2) \times (3 \times 10^8) \text{ ms}^{-1}} = 3.85 \mu\text{s}$$

$$\begin{aligned} \text{Delay in reflected signal w.r.t. direct signal} &= t_{\text{indirect}} - t_{\text{direct}} \\ &= 3.85 - 3.33 \mu\text{s} = 0.52 \mu\text{s} \end{aligned}$$



Radio Band Qualities

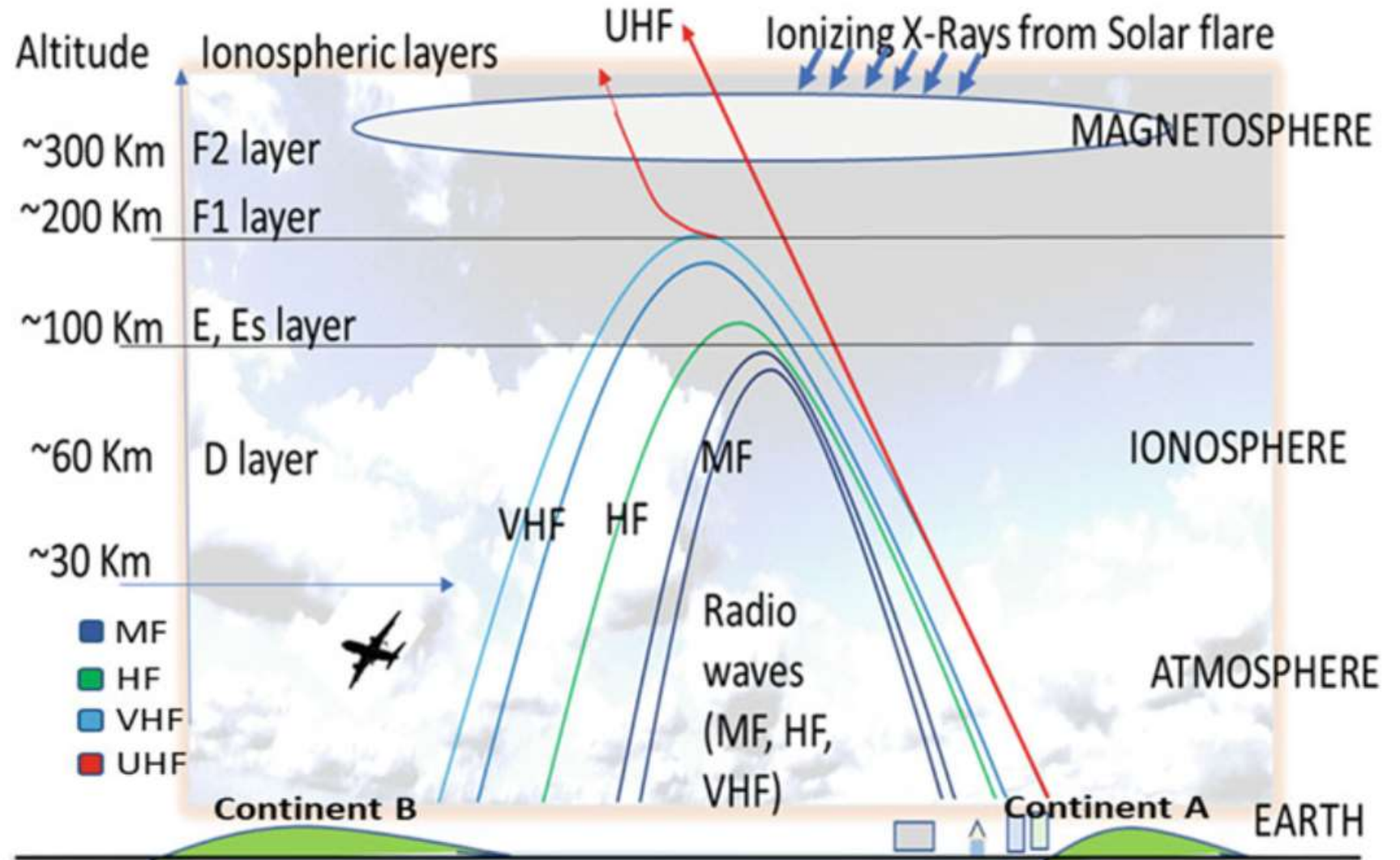
❑ Radio band is separated from other bands due to specific qualities of waves within this band such as.

1. Radio waves can travel over great distances “without the need for a metallic wire”.
2. Most objects except electrical conductors such as metals, carbon, etc., do not stop these waves. Therefore, propagation through natural impediments such as trees, layers of insulation, and other objects is possible (but signal strength reduces). Therefore, only metal as well as electrical conductors are considered “blockages”.
3. The largest user of “reflected waves” is the broadcast industry. Short waves are radio waves reflected off different layers of ionosphere.
4. It is possible to retain orientation of the electric field waves based on the physical orientation of the source antenna (vertical or horizontal).
5. Radio waves can carry information effectively over great distances—this allows a transmitter to be placed on a satellite, a space probe or even in Voyager that has moved beyond our solar system. It is possible to carry on effective communication independent of the distance. System design incorporates distance into calculations.

Reflection of Radio Waves—“Ionosphere” Reflector in the Sky

Let us now consider the Ionosphere which refers to different layers above the earth, that reflect different frequency ranges of radio waves. This property of the Ionosphere in the sky has been used for almost one hundred years. Figure 2.3 depicts an imaginary view of the earth involving two separate continents. It allows radio broadcast and other communication services over long distances. This service is commonly referred to as “short wave radio” and allows communication beyond the horizon. This can be used in various applications that need communication over very long distances (hundreds of km).

Fig. 2.3 Ionospheric layers and radio waves



- ❑ There is a point when finally, in the upper UHF band these waves penetrate the ionosphere and travel up towards outer space. Note that all these bands are within the radio frequency range.
- ❑ Medium Frequency MF (300 kHz–3 MHz) is used by AM radio, amateur radio and avalanche beacons.
- ❑ H.F. (High Frequency 3 to 30MHz) Long Range communications. Shipping, Aircraft, World Broadcast Communications, Radio Amateurs. Use involves reflecting the signal off the ionosphere back down to waiting receiving stations. Prone to atmospheric changes causing fading and noise. Range from 500 to thousands of Kilometres.
- ❑ V.H.F. (Very High Frequency 30 - 300 MHz) Medium range communications. Fleet vehicles, mobile, coastal shipping and air to tower communications. Range 70-100km (aircraft several hundred km).
- ❑ U.H.F (Ultra High Frequency 300-3000 MHz) This is the domain of such things as Police handheld radios, cell-phones, T.V., spacecraft to ground communications, and space/satellite communication. In the high U.H.F. range the signal can "bounce" off buildings and reflect until it is detected by a receiver.

Atmospheric windows to EM radiation

- ❑ Earth's atmosphere acts as an opaque barrier to much of the electromagnetic spectrum.
- ❑ The atmosphere absorbs most of the wavelengths shorter than ultraviolet, most of the wavelengths between infrared and microwaves, and most of the longest radio waves.
- ❑ For radio astronomers this leaves only short wave radio to penetrate the atmosphere and bring information about the universe to our Earth-bound instruments.
- ❑ The main frequency ranges allowed to pass through the atmosphere are referred to as the radio window.
- ❑ The radio window consists of frequencies which range from about 5 MHz (5 million hertz) to 30 GHz (30 billion hertz).
- ❑ The low-frequency end of the window is limited by signals being reflected by the ionosphere back into space, while the upper limit is caused by absorption of the radio waves by water vapor and carbon dioxide in the atmosphere.
- ❑ As atmospheric conditions change the radio window can expand or shrink. On clear days with perfect conditions signals as high as 300GHz have been detected.

Penetrates
Earth
Atmosphere?



Wavelength
(meters)



About the size of...



Buildings



Humans



Honey Bee



Pinpoint



Protozoans



Molecules

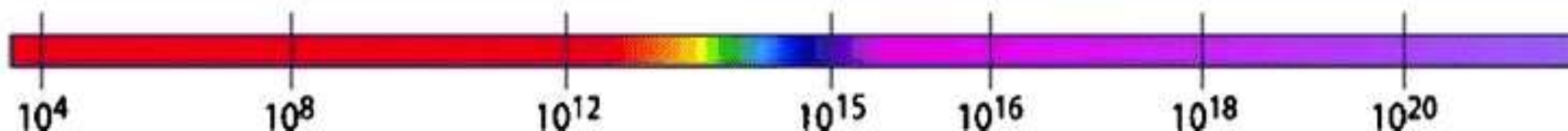


Atoms



Atomic Nuclei

Frequency
(Hz)



Wireless System Types :

- 1. Cellular System:** Cellular system gained popularity due to a variety of reasons. Primary among them is the ability of cell phone to be used anywhere including moving at considerable speeds. This is not the only reason. There are others, such as ability to maintain a consistent set of services that users have subscribed to.

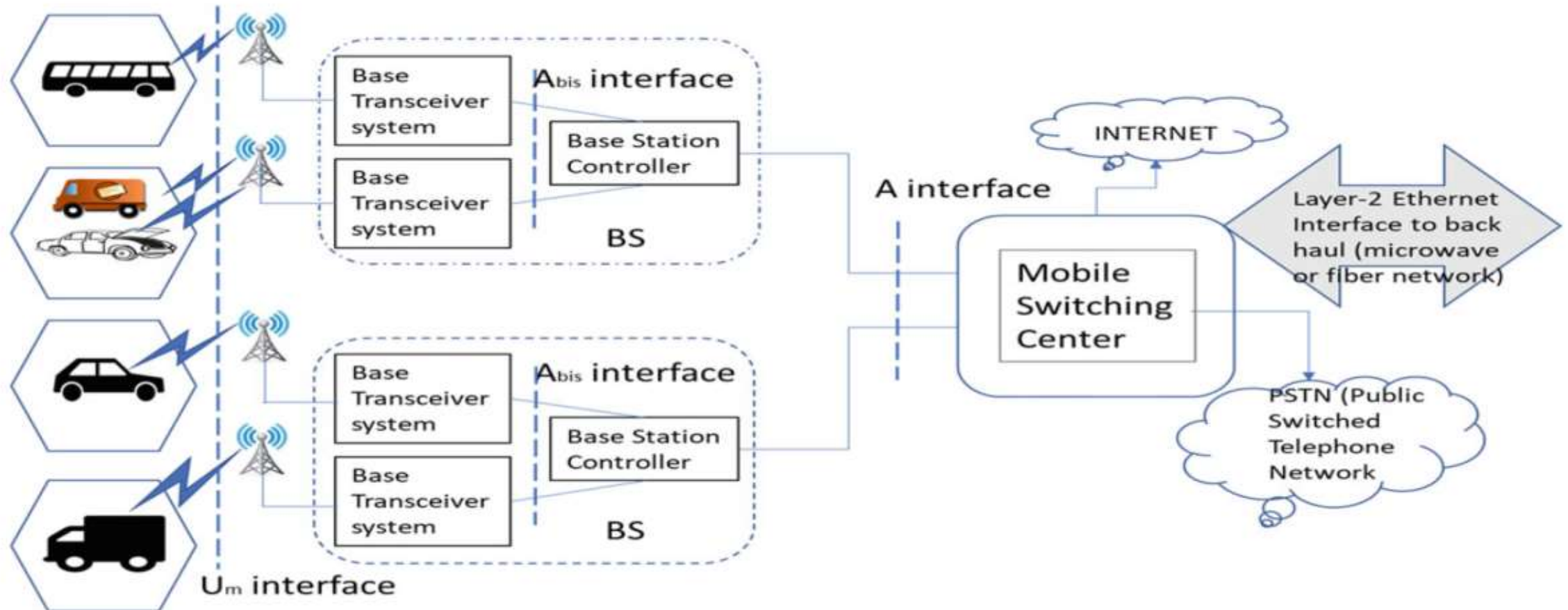
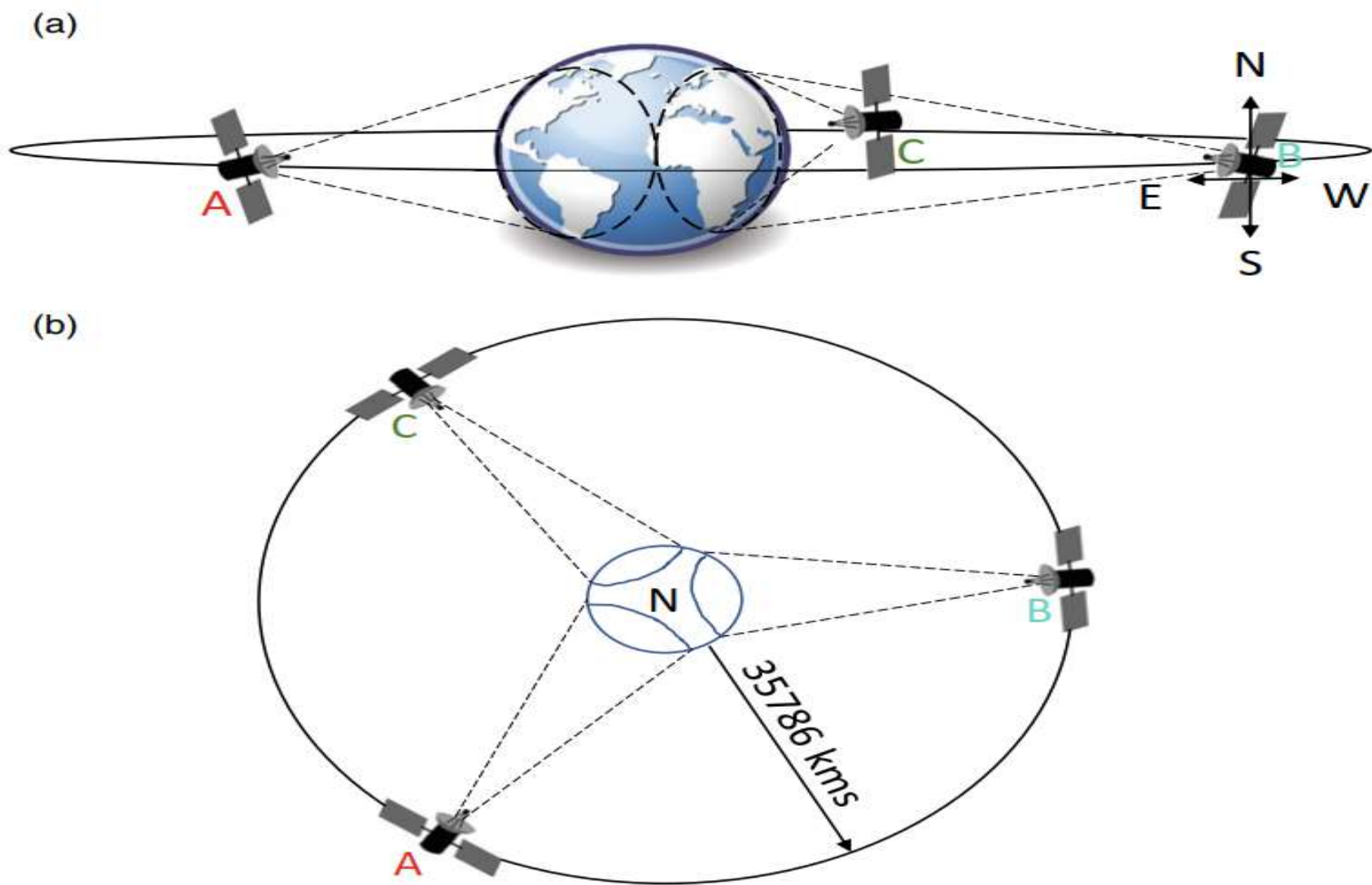


Fig. 3.1 General overview of cellular network architecture

2. Satellite System:

Fig. 3.3 **a** Geostationary Orbit—
viewed from equatorial plane.
b Geostationary orbit viewed
from above (North Pole)



3. Wireless LAN (WLAN)

Wireless LAN (Local Area Network) is typically a 100 m extension of the wired LAN—almost always within a building or campus. However, its widespread use as WiFi has made it very popular as a destination wireless link at home, in office, at rail stations, airports, etc. It is perhaps one of the most popular of wireless technologies in urban areas where Local Area Network (LAN) and fiber network normally exists. WLAN is a systematic extension of this network to provide flexibility to the end user. This topic includes a variety of Personal Area Networks (PAN) most are based on IEEE standards. Many such as Bluetooth, ZigBee, WiMax, and others, are in this category.

What is Mobile Computing

Mobile Computing is an umbrella term used to describe technologies that enable people to access network services anyplace, anytime, and anywhere.

Mobile Computing Applications

- Stock Information Collection/Control
- Emergency services
- For Estate Agents
- In courts
- In companies
- Credit Card Verification
- Taxi/Truck Dispatch
- Electronic Mail/Paging
- Medical

Nurses/Doctors in Medical offices are now using Wireless Tablet PCs/WLAN to collect and share patient information.

➤ Vehicles

- transmission of news, road condition, weather, music via DAB
- personal communication using GSM
- position via GPS
- local ad-hoc network with vehicles close-by to prevent accidents, guidance system, redundancy
- vehicle data (e.g., from busses, high-speed trains) can be transmitted in advance for maintenance

➤ Sales

Sales representatives are using Tablet PCs with Smart phones for presentation, transmitting/access information among office, hotel, and customer location.

➤ Emergencies

- Early transmission of patient data to the hospital, current status, first diagnosis
- Provide mobile infrastructure in dealing with Natural Disaster (earthquake, hurricane, fire), terrorist attacks, war, ...

Characteristics of Mobile Computing

- ❑ **Ubiquity** – The ability of a user to perform computation from anywhere and at anytime.(present everywhere)
- ❑ **Location awareness** – Many applications requires or value additions by location based services.(GPS)
- ❑ **Adaptation-** Ability to adjust to bandwidth fluctuations without inconveniencing the user.
- ❑ **Broadcast-** Efficient delivery of data can be made simultaneously to hundreds of mobile users
- ❑ **Personalization** – Services in a mobile environment can be easily personalized according to a user's profile.